

A unified and simplified treatment of the non-linear equilibrium problem of double-row rolling bearings.

Part 2: application to taper rolling bearings supporting a flexible shaft

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Abstract: A general theoretical model to analyse the internal interactions within various types of double-row rolling bearing was presented in Part 1 of this two-part paper. The performance of the rolling bearing model is shown here for a bearings–flexible-shaft system. Double-row tapered roller bearings (in a back-to-back or face-to-face arrangement) were chosen as support bearings. Investigations of the effect of shaft mounting on support bearings (cantilever or straddle arrangement) and on the effect of preload are presented and discussed. It is shown that, when an optimum value of preload or fatigue life of the bearing arrangement is expected, shaft bending cannot be ignored.

Keywords: double-row tapered roller bearing, flexible shaft, straddle and cantilever mounting, fatigue life

NOTATION

c	distance to the load point from the left-hand bearing centre (m)
d_{ech}	shaft equivalent diameter (m)
DB	back-to-back arrangement
DF	face-to-face arrangement
DRTRB	double-row tapered roller bearing
E	modulus of elasticity (N/m)
f	shaft deflection (m)
F_a	axial force applied on the shaft (N)
F_r	radial force applied on the shaft (N)
F_x	axial force induced in the bearing (N)
F_y	radial force induced in the bearing (N)
I	cross-section moment of inertia (m ⁴)
l	distance between bearing centres (m)
l_t	shaft length that affects the stiffness (m)
L_{10}	bearing arrangement life for 90 per cent reliability (10 ⁶ rev)

M_s	shaft bending moment (N m)
M_z	bearing moment load around the Z axis (N m)
TRB	tapered roller bearing
δ	bearing radial deflection (m)
δ_p	axial clearance/compression (m)
θ	bearing angular deflection (m)

Subscripts

a	axial
A	bearing A
B	bearing B
i	i th region of the shaft
r	radial

1 INTRODUCTION

In Part 1 [1] of this two-part paper, a general mathematical model to describe complex mechanical interactions found within various types of double-row rolling bearings was presented. In many engineering applications these bearings are used to obtain maximum

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rigidity of the shaft-support bearings system. Thus, spiral bevel gears, fans, blowers, centrifugal pumps, machine tools, advanced transmissions and other precision machinery illustrate situations in which a rotating shaft must be precisely located with respect to its housing. In addition, in these situations, the bearings must be preloaded to remove all internal clearance and to optimize the fatigue life. The bearing arrangement of rotating machine components, e.g. a shaft, generally requires two bearings to support and locate the radial and axial components relative to the stationary part of the machine, such as the housing. It is considered that high rigidity of each part of the system is generally indispensable to achieve high accuracy of the shaft rotation. In particular, the deflection of the bearing–shaft system affects the working precision. As early as 1973, Andreason [2] pointed out the need for computing the complete load support system. While developing solutions for critical rolling element applications, such as multiple-shaft gear drives, the designer is often hampered by incomplete knowledge of the effect of mutual deflection of all components in the system. Bhardwaj *et al.* [3] have experimentally studied a complete multiple-shaft gear drive system (equipped with single-row roller bearings), using the contact pattern analysis technique. This approach has shown several anomalies in the load distribution among rollers when rings are misaligned due to the shaft bending. The constraints imposed on the space and mass for example affect the load bearing support stiffness and thus require an accurate forecast of the resulting load and displacement vector distribution at the gear mesh. An analysis and series of computations were carried out by Gassel and Pirvics [4] to explore the competing prototype design concepts of a shaft and tapered roller bearing (single-row) system to support the input pinion of an advanced commercial helicopter transmission. Finally, Yamada *et al.* [5] investigated the influence of the distance between bearing centres upon the stiffness.

The purpose of this paper is to investigate the stiffness of double-row rolling bearings–shaft systems. Firstly, the governing equation describing the shaft bending and consequently the reaction loads that depend on the bearing displacements (translations and rotations) will be presented. Secondly, the model will be applied to a typical rolling bearings–shaft assembly. The support bearings chosen were double-row tapered roller bearings (DRTRBs) in a back-to-back (DB) and a face-to-face (DF) arrangement.

In modern engineering applications, designers are asked to introduce rolling bearings into the modelling of the mechanical system in order to determine accurately the shaft deflections. The shaft-support bearing model and associated numerical simulations presented here for DRTRBs are well adapted to static external loads. The corresponding static model consists of a set of non-

linear algebraic equations solved by a standard iterative method.

2 BEARING DESCRIPTION

The use of DRTRBs is increasing, e.g. for automotive (front-wheel), railway (high-speed train and locomotive axle journal, mini and metro rear wheels) or rolling mill (back-up roll) applications. This is due to the bearing's ability to support large combined loads for a given envelope and mass. DRTRBs are used for their high load-carrying capacity, load in both axial directions being permitted, and also since the bearing width is smaller than in the case of a set of two single-row tapered roller bearings (TRBs). The load lines of a DRTRB (DB) diverge towards the bearing axis. Therefore such bearing arrangements are relatively stiff and can accommodate tilting moments, while the load lines of a DRTRB (DF) converge towards the bearing axis (Fig. 1), giving it more ability to rotate under moment loading.

The effective spread E is a function of the bearing arrangement and contact angle. A wider effective spread usually acts to reduce the load on rolling element. However, under certain loading conditions such as high bending moment the rolling element load may increase with increasing effective spread.

3 SHAFT-BEARING LOADS AND DEFLECTIONS

When looking for a high stiffness, it is necessary to take into account both rolling bearing stiffness and shaft bending. The system is statically indeterminate and the static equilibrium equations are insufficient to solve the bearing reactions. The complexity of the problem is due to the bending of the shaft that affects the bearings loading [6]. The model presented in Part 1 of this two-part paper was used to evaluate some design aspects for both a straddle arrangement (where the radial load is between bearings) and a cantilever arrangement (where the radial load is outboard of the two bearings), as shown in Figs 2 and 3. The shaft is free to float axially

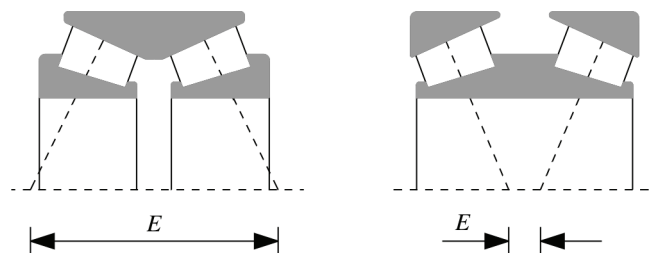


Fig. 1 Effective spread E of a DRTRB (DB) (left) and a DRTRB (DF) (right)

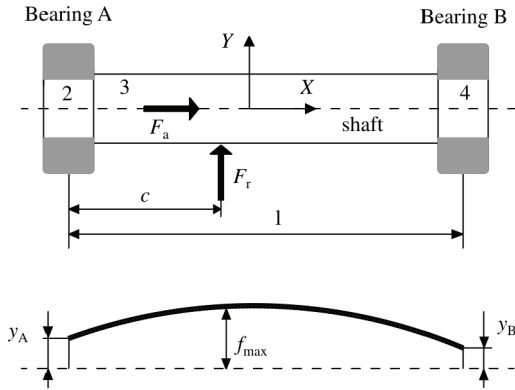


Fig. 2 Geometry of the straddle arrangement

and therefore bending it does not induce a supplementary thrust load.

The external loading applied to the shaft is balanced by rolling element loads. The radial load applied on the shaft, F_r , produces in the support bearings radial loads F_{yA} and F_{yB} and moment loads M_{zA} and M_{zB} . An additional axial load F_a applied on the shaft may be also considered in the present analysis. As a result, the inner ring of the support bearings will move in the radial direction by δ_{yA} and δ_{yB} and will be misaligned by θ_{zA} and θ_{zB} . The reaction loads on the shaft to the bearings, F_{yA} , F_{yB} , M_{zA} and M_{zB} , should be expressed as functions of F_r , δ_{yA} , δ_{yB} , θ_{zA} , θ_{zB} , support system geometry and shaft material constants. Finally the reaction loads are deduced starting from the governing equation describing the shaft bending using simple beam theory. Thus, in the XY plane, the bending moment at any section is defined as

$$\frac{d^2 y}{dx^2} = -\frac{M_s(x)}{EI(x)} \quad (1)$$

in which

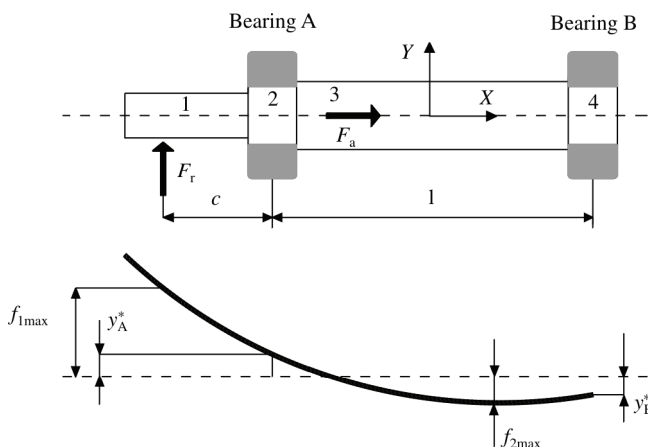


Fig. 3 Geometry of the cantilever arrangement

E = elasticity modulus of the shaft
 M_s = bending moment of the shaft
 I = shaft cross-section moment of inertia

Integrating equation (1) twice for two distinct regions, $x \in [0, c]$ and $x \in [c, l]$, and combining the resulting expressions, it yields the following equations for reaction loads and shaft deflection (at any cross-section).

Firstly, the straddle design equations are

$$F_{yA} = \frac{F_r(l-c)^2(l+2c)}{l^3} - \frac{6EI}{l^2} \times \left(\theta_{zA} + \theta_{zB} + 2 \frac{\delta_{yA} - \delta_{yB}}{l} \right) \quad (2a)$$

$$F_{yB} = \frac{F_r c^2(3l-2c)}{l^3} + \frac{6EI}{l^2} \times \left(\theta_{zA} + \theta_{zB} + 2 \frac{\delta_{yA} - \delta_{yB}}{l} \right) \quad (2b)$$

$$M_{zA} = \frac{F_r c(l-c)^2}{l^2} - \frac{2EI}{l} \times \left(2\theta_{zA} + \theta_{zB} + 3 \frac{\delta_{yA} - \delta_{yB}}{l} \right) \quad (3a)$$

$$M_{zB} = \frac{F_r c^2(l-c)}{l^2} + \frac{2EI}{l} \times \left(\theta_{zA} + 2\theta_{zB} + 3 \frac{\delta_{yA} - \delta_{yB}}{l} \right) \quad (3b)$$

Also, for $x \in [0, c]$,

$$f(x) = \frac{1}{EI} \left[-\frac{F_{yA} x^3}{6} + \frac{M_{zA} x^2}{2} + EI(\delta_{yA} + x\theta_{zA}) \right] \quad (4a)$$

and, for $x \in [c, l]$,

$$f(x) = \frac{1}{EI} \left\{ -\frac{F_{yA}}{6} [l^2(2l-3x) + x^3] + \frac{M_{zA}}{2} (l-x)^2 + \frac{F_y}{6} \times [x^2(x-3c) - l^2(3x+3c-2l) + 6lcx] + EI[\delta_{yB} - \theta_{zB}(l-x)] \right\} \quad (4b)$$

Secondly, the cantilever design equations are

$$F_{yA} = F_r - \frac{6EI}{l^2} \left(\theta_{zA} + \theta_{zB} + 2 \frac{\delta_{yA} - \delta_{yB}}{l} \right) \quad (5a)$$

$$F_{yB} = \frac{6EI}{l^2} \left(\theta_{zA} + \theta_{zB} + 2 \frac{\delta_{yA} - \delta_{yB}}{l} \right) \quad (5b)$$

$$M_{zA} = F_r c + \frac{2EI}{l} \left(2\theta_{zA} + \theta_{zB} + 3 \frac{\delta_{yA} - \delta_{yB}}{l} \right) \quad (6a)$$

$$M_{zB} = \frac{2EI}{l} \left(\theta_{zA} + 2\theta_{zB} + 3 \frac{\delta_{yA} - \delta_{yB}}{l} \right) \quad (6b)$$

Also, for $x \in [0, c]$,

$$f(x) = \frac{1}{EI} \left\{ \frac{F_r}{6} (x^3 - 3c^2x + 2c^3) + EI[\delta_{yA} + \theta_{zA}(x - c)] \right\} \quad (7a)$$

and, for $x \in [c, l]$

$$f(x) = \frac{1}{EI} \left\{ \frac{F_r}{6} [x^3 - 3x(c+l)^2 + 2(c+l)^3] - \frac{F_{yA}}{6} x^2(x-3c) + \frac{F_{yA}}{6} (c+l)^2(c-2l) - \frac{F_{yA}}{2} x(c^2 - l^2) - \frac{M_{zA}}{2} [x - (c+l)]^2 + EI[\theta_{zB}(x - c - l) + \delta_{yB}] \right\} \quad (7b)$$

In equations (4) and (7), x represents the distance along the shaft and f the shaft deflection in the direction perpendicular to x . If the shaft is very flexible and the bearings are rigid with regard to misalignment, then θ_{zA} and θ_{zB} are approximately zero. As illustrated in Figs 2 and 3, the shaft contains some distinct regions, numbered from 1 to 4. The diameter and active length of each shaft region are denoted by l_i and d_i respectively, with $i = 1, \dots, 4$. With these specifications the shaft moment of inertia can be calculated as

$$I = \frac{\pi d_{ech}^4}{64} \quad (8)$$

with d_{ech} the equivalent diameter of the shaft, expressed by the relation

$$d_{ech} = \sqrt[4]{\frac{l_t}{\sum_i l_i/d_i^4}} \quad (9)$$

where l_i is the active length of section i of the shaft [e.g., at section 1 of the cantilever design (Fig. 3), $l_1 = c - 0.5l_2$], l_t is the shaft length that affects the stiffness behaviour (for the straddle arrangement, $l_t = 1$ and, for the cantilever arrangement, $l_t = c + l$), with l the distance between bearing centres [when the support bearings are single-row tapered roller bearings (SRTRBs), then l is the distance between the pressure cone apexes of the bearings] and, for the cantilever design, c is the distance between the loading point and the closest bearing centre (for SRTRBs and the configuration in Fig. 3, c is the distance between the loading point and the pressure cone apex of the left-hand-side bearing).

Taking into account the active length of the shaft sections that affect the stiffness behaviour, equation (9) is as follows. In the case of the straddle design (see

Fig. 2),

$$d_{ech} = \sqrt[4]{\frac{l_t}{0.5l_2/d_2^4 + l_3/d_3^4 + 0.5l_4/d_4^4}}$$

where $l_t = 1 = 0.5l_2 + l_3 + 0.5l_4$. In the case of the cantilever design (see Fig. 3),

$$d_{ech} = \sqrt[4]{\frac{l_t}{(c - 0.5l_2)/d_1^4 + l_2/d_2^4 + l_3/d_3^4 + 0.5l_4/d_4^4}}$$

where $l_t = c + 1 = c + 0.5l_2 + l_3 + 0.5l_4$.

4 BEARING SUPPORT EQUILIBRIUM

The bearing-flexible shaft assembly represents a statically indeterminate system. In our case, when the shaft is loaded axially by F_a and radially by F_r , there are five unknowns: δ_{xA} , δ_{xB} , δ_{yB} , θ_{zA} and θ_{zB} . In the axial direction, the shaft is considered to be infinitely rigid and consequently the axial deflections δ_x are assumed to be the same for all support bearings. To find the unknowns mentioned above, it is necessary to solve the following equilibrium system for support bearings, written in a concentrated vectorial form as

$$(\mathbf{F}_{shaft})_{A,B} + (\mathbf{F}_{bearing})_{A,B} = \mathbf{0} \quad (10)$$

where $\mathbf{F}_{shaft}$ is the vector of loads applied on the bearing by the shaft. This vector can be interpreted as the external loads for the bearing; $\mathbf{F}_{shaft} = \{F_x, F_{yA,B}, M_{zA,B}\}^T$. $\mathbf{F}_{bearing}$ denotes the vector of reaction loads and moments within the bearing. The subscripts A and B refer to the support bearings A and B respectively (Figs 2 and 3).

The $\mathbf{F}_{shaft}$ vector components are defined in equations (2) and (3) for the straddle arrangement and in equations (5) and (6) for the cantilever arrangement. The components of the vector $\mathbf{F}_{bearing}$ are defined by equation (27) of Part 1 [1] for bearings having point contacts and by equation (31) of Part 1 [1] for bearings having line contacts. The system of equations represented in equation (10) is solved using the Newton-Raphson iterative algorithm.

5 CALCULATION EXAMPLES AND DISCUSSION

Numerical results of bearings-shaft computer simulations have shown that both stiffness and shaft response can be significantly affected by a variety of data such as the bearing type, bearing arrangement, preload or the external loading. The calculation examples were performed for two comparable structural assemblies: firstly, a cantilever arrangement and, secondly, a straddle

arrangement. For each design configuration the DRTRB type was chosen, both in a DB arrangement and in a DF arrangement. The cantilever and straddle geometries are given in Table 1. The discussion on the simulation results is focused on bearing stiffness and L_{10} fatigue life. The L_{10} fatigue life for the bearings–shaft assembly is described in the Appendix.

The relevant dimensions of the chosen bearing are listed in Table 2, with identical geometry for each row.

For each design configuration (straddle and cantilever), the support bearings (left and right) are of the same type [DRTRB (DB) or DRTRB (DF)]. For each numerical simulation the axial load F_a has been chosen to be identical with the radial load F_r , varying in the range 3000–9000 N. The total initial axial clearance/compression varies from -0.4 to 0.4 mm, leading to an axial compression per bearing row, δ_p , in the range from -0.1 to 0.1 mm.

5.1 Results for straddle design

The effect of distance between bearing centres on the maximum shaft deflection $f_{\max}$ for axial and radial loads of 3000, 6000 and 9000 N (applied on the shaft) is shown in Fig. 4. In all these calculations the axial compression is $\delta_p = 0.02$ mm per row or 0.08 mm on the whole system. It can be observed that the stiffness of the supporting TRB in the DB arrangement is higher than for the TRB in the DF arrangement.

A series of curves giving the maximum shaft deflection, for the same shaft loads as those used in Fig. 2, and for various axial compression ranging from 0 to 0.1 mm per row or from 0 to 0.4 mm for the whole bearings–shaft system is presented in Fig. 5. In this case the distance between supporting bearings was chosen to be equal to $l = 354$ mm. It can be observed that, when the axial compression δ_p exceeds the value of 0.04 mm, the shaft deflection remains approximately constant for both the TRB in the DB arrangement and the TRB in the DF arrangement. From Figs 4 and 5 it can be observed that the DB or DF configuration affects the amount of shaft deflection. This is due to different abilities of the DB and DF arrangements [the effective spreads E for DB and DF arrangements are different (see Fig. 1) and therefore have different influences on the bearing stiffness] to support the bending moments

generated by the external radial load. The angular stiffness (due to the moment load) of the DB arrangement with larger spread E is higher than that of the DF arrangement with smaller spread. This result is consistent with ball bearing arrangement behaviour, where the ‘O’-type assembly is known to be stiffer than the ‘X’-type assembly (in terms of angular stiffness due to the moment load).

Comparative results concerning the axial and radial bearing loads generated by the external loading applied to the shaft in straddle design for DRTRBs (DB) and DRTRBs (DF) are listed in Table 3. The shaft is loaded axially with 3000 N and radially with 3000, 6000 and 9000 N. The radial load acts at the middle of the shaft ($c = 0.5l$). Bearings A and B are the left- and right-hand bearings respectively (see Fig. 2). The positive or negative values correspond to the positive or negative sense of the bearing inertial frame. It can be noted that, when the shaft is assumed to be infinitely rigid, the induced loads do not depend on the bearing arrangement (DB or DF). The differences can only be seen between the bearing rows (left- or right-hand row) by the action of the axial external load. Moreover, for $c = 0.5l$ both the axial and the radial loads applied on the shaft are equally distributed between the two support bearings (this distribution can be altered when $c \neq 0.5l$).

Figures 6a and b show the variation in the L_{10} fatigue life as functions of the axial compression for two shaft loads of 3000 and 6000 N ($= F_a = F_r$). The initial axial clearance/compression per bearing row varies in the range between -0.1 and $+0.1$ mm. A comparison between the L_{10} fatigue lives of the support bearing assembly calculated for flexible and rigid shafts is also made in the same figure.

It can be observed both in Fig. 6a and in Fig. 6b that, for low values of axial clearance, the L_{10} life for the DB arrangement exceeds that for the DF arrangement. The situation is reversed when the axial compression exceeds a certain value depending on the shaft loading. Thus, the optimum axial clearance/compression must be correctly chosen as a compromise between the L_{10} life and the

Table 1 Dimensions of straddle and cantilever designs

l_1	46 mm	d_1	30 mm
l_2	34 mm	d_2	35 mm
l_3	From 100 to 400 mm	d_3	40 mm
l_4	34 mm	d_4	35 mm
l	From 134 to 434 mm		
c	80 mm, cantilever design		
c	0.5 <i>l</i> , straddle design		

Table 2 Geometrical parameters of DRTRBs

Symbol	Units	Value
Pitch diameter, d_m	mm	53.5
Roller mean diameter, D	mm	9
Roller length, L	mm	12
Roller chamfer, r_1, r_2	mm	0.5
Half included cup angle, α	deg	14.036
Half included cone angle, β	deg	10.036
Flange angle, ν	deg	10.667
Half included roller angle, ε	deg	2
Number of rollers, Z	—	20
Distance between rows, e^*	mm	10.2

*The parameter e represents the axial distance between bearing row centres.

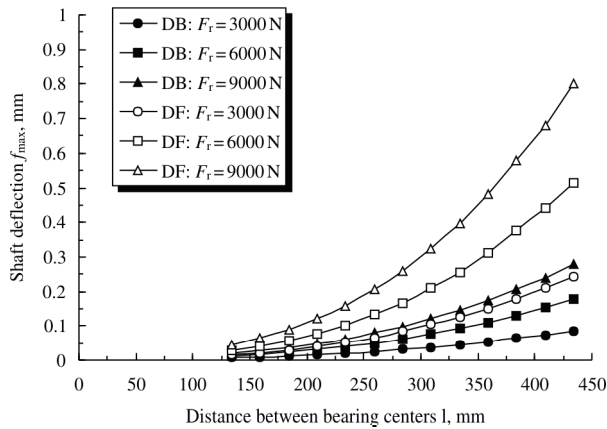


Fig. 4 Maximum shaft deflection versus distance between bearing centres ($\delta_p = 0.02$ mm)

stiffness of the bearings–shaft assembly. Also, it can be observed that there is a significant difference between the optimum values of axial clearance/compression for the case of flexible and rigid shafts.

However it is worth noting that, in the simulations presented here, roller profiles have been assumed to be rectilinear since industrial and realistic data are not available. In addition, roller (and raceway) profiles vary usually from one manufacturer to another. They vary also in time depending on the manufacturing process. It is clear that the use of another roller profile such as a crowned or a logarithmic profile might lead to a different fatigue life by reducing the stress concentration at the roller end or raceway contacts, as shown for example by Gohar and co-workers for dry [7] or lubricated contacts [8]. The stiffness of individual DRTRB or the DRTRB arrangement should be less affected.

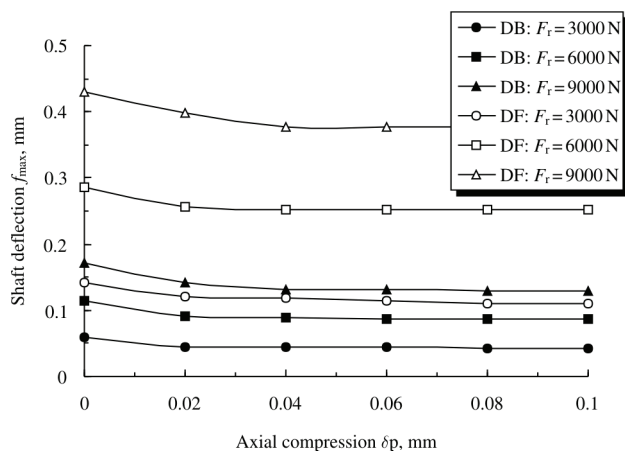


Fig. 5 Maximum shaft deflection versus axial compression ($\delta_p = 0.02$ mm and $l = 354$ mm)

5.2 Results for cantilever design

For cantilever design, the shaft bending can be analysed in two regions: $[0, c]$ and $[c, 1]$. In each of these regions, the shaft is characterized by a maximum deflection: $f_{1\max}$ at $x=0$ and $f_{2\max}$ for $x \in [c, 1]$ (see Fig. 3). The influence of the distance between the bearing centres on the maximum shaft deflection is illustrated in Figs 7a and b for three loads applied on the shaft: 3000, 6000 and 9000 N. These results show once again that the stiffness in the case of the support bearings of type DRTRB in the DB arrangement is higher than in the DF arrangement.

For the same shaft loading as that used in Figs 7a and b, the evolution of the maximum shaft deflection as function of the axial compression varying from 0 to 0.1 mm per row or from 0 to 0.4 mm for the whole bearings–shaft system is presented in Figs 8a and b. Similarly to Fig. 5, it can be noted that, when the axial compression δ_p exceeds the value of 0.04 mm, the stiffness variation remains approximately constant for both bearing types (DRTRBs in the DB arrangement and DRTRBs in the DF arrangement).

Comparative results concerning the axial and radial bearing loads on the support bearings [DRTRBs (DB) and DRTRBs (DF)] for cantilever design are listed in Table 4. The loading conditions are similar to those used in Table 3. The significant difference observed between the results of calculations performed for flexible and for infinitely rigid shafts should be mentioned. The type of bearing affects also significantly the results of the simulation (DB or DF arrangement).

Figures 9a and b present the relation between the L_{10} fatigue life and the axial clearance/compression for two shaft applied loads (3000 and 6000 N). The axial clearance/compression per bearing row varies from -0.1 to 0.1 mm. In addition a comparison between the L_{10} fatigue lives of the support bearings assembly calculated for flexible and rigid shafts is also made.

The maximum values of the L_{10} life are more pronounced in the DF arrangement than in the DB arrangement. This is why the DF arrangement may be preferred for cantilever design. Again it should be mentioned that an improvement in the fatigue life may be observed, both in the simulation and in real applications, when considering an optimized roller profile instead of the rectilinear profile used here as input data.

6 CONCLUSIONS

The theoretical model presented in Part 1 of this two-part paper and developed for various types of double-row rolling bearing was used to study the performances, for both a straddle mounting (where the shaft load is located axially between the two bearings) and a

Table 3 Comparison between bearing loads calculated for flexible and rigid shafts in the straddle design ($\delta_p = 0.02$ mm, $F_a = 3000$ N, $l = 354$ mm and $c = 0.5$ l)

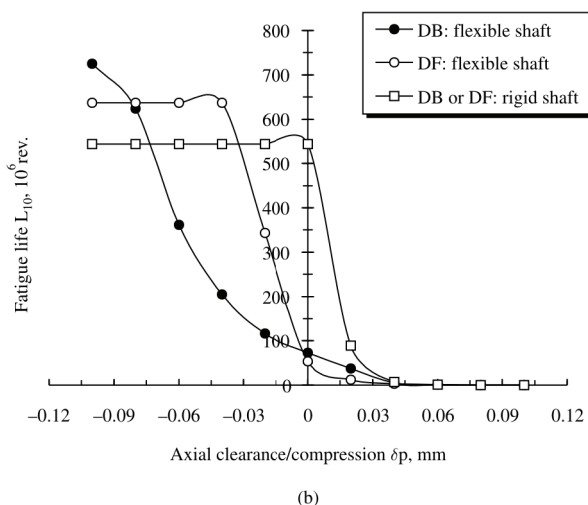
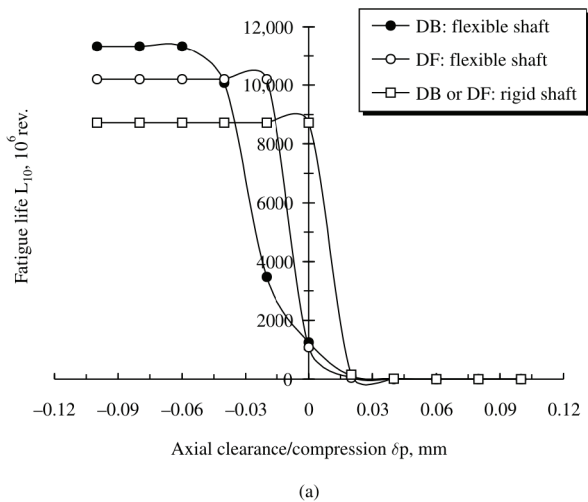
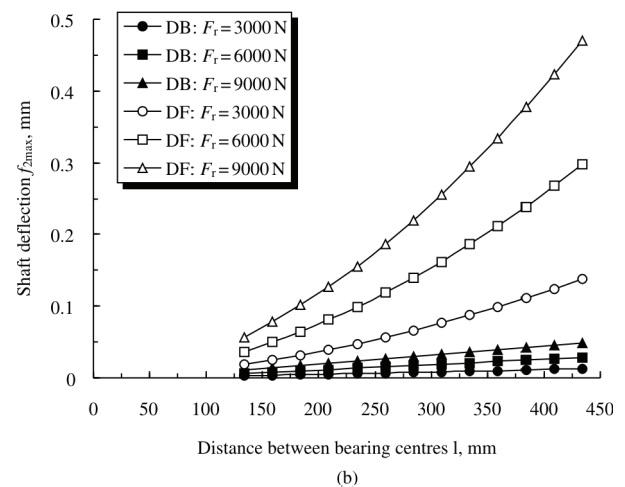
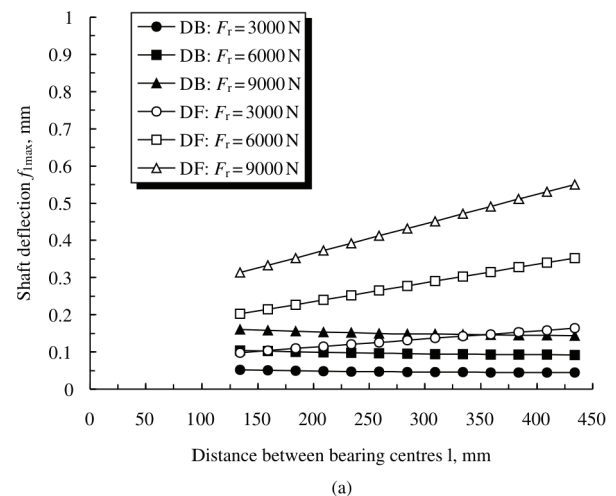
F_r (N)	DRTRB (DB)*				DRTRB (DF)*				DRTRB†	
	Bearing A		Bearing B		Bearing A		Bearing B		Bearings A, B	
	F_{xA} (N)	F_{yA} (N)	F_{xB} (N)	F_{yB} (N)	F_{xA} (N)	F_{yA} (N)	F_{xB} (N)	F_{yB} (N)	$F_{xA,B}$ (N)	$F_{yA,B}$ (N)
3000	1483	1499	1517	1501	1542	1496	1458	1504	1500	1500
6000	1356	2990	1644	3010	1957	3042	1043	2958	1500	3000
9000	826	4464	2170	4536	2532	4566	468	4434	1500	4500

*Considering that the shaft bends.

†Considering that the shaft is infinitely rigid.

cantilever mounting (where the shaft load is outboard of the bearings). DRTRBs in DB and DF arrangements were chosen as support bearings. The effect of the shaft bending was included in the analysis. The influences of

varying parameters including the shaft applied load, the axial clearance/compression (or preload) and the load support geometry (distance between bearing centres) were investigated. Results show that the shaft bending

**Fig. 6** L_{10} fatigue life versus axial clearance/compression: (a) $F_a = F_r = 3000$ N; (b) $F_a = F_r = 6000$ N**Fig. 7** Maximum shaft deflection versus distance between bearing centres ($\delta_p = 0.02$ mm and $c = 80$ mm): (a) variation in $f_{1\max}$; (b) variation in $f_{2\max}$

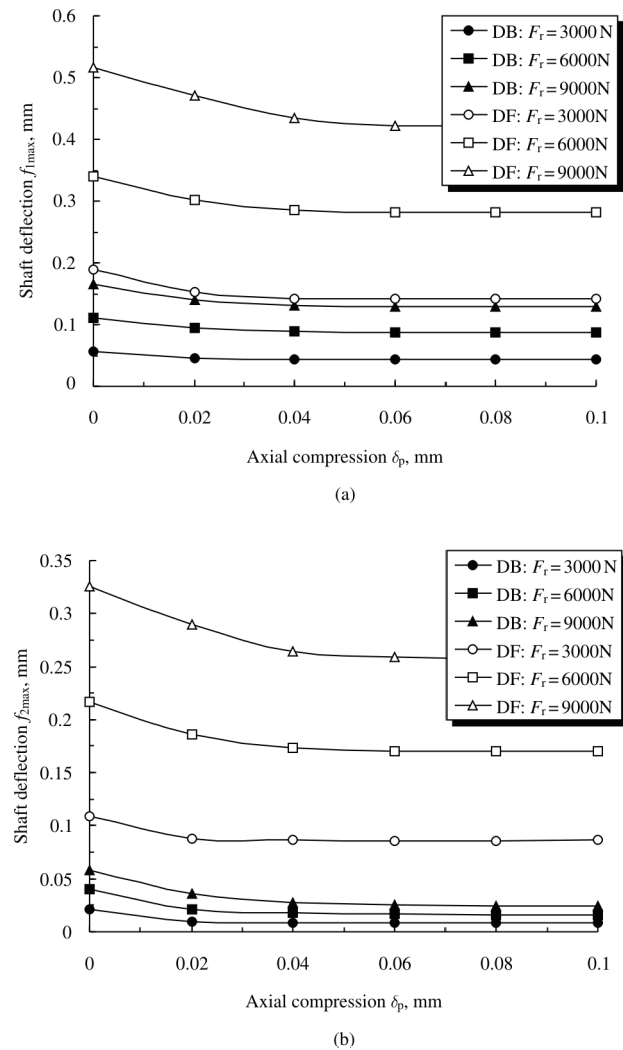


Fig. 8 Maximum shaft deflection versus axial compression ($l=354\text{ mm}$): (a) variation in $f_{1\text{max}}$; (b) variation in $f_{2\text{max}}$

must not be ignored in the bearings–shaft system when high rigidity is expected. The sensitivity of preload on the design performance (such as the L_{10} fatigue life) was also investigated. The optimum axial clearance/com-

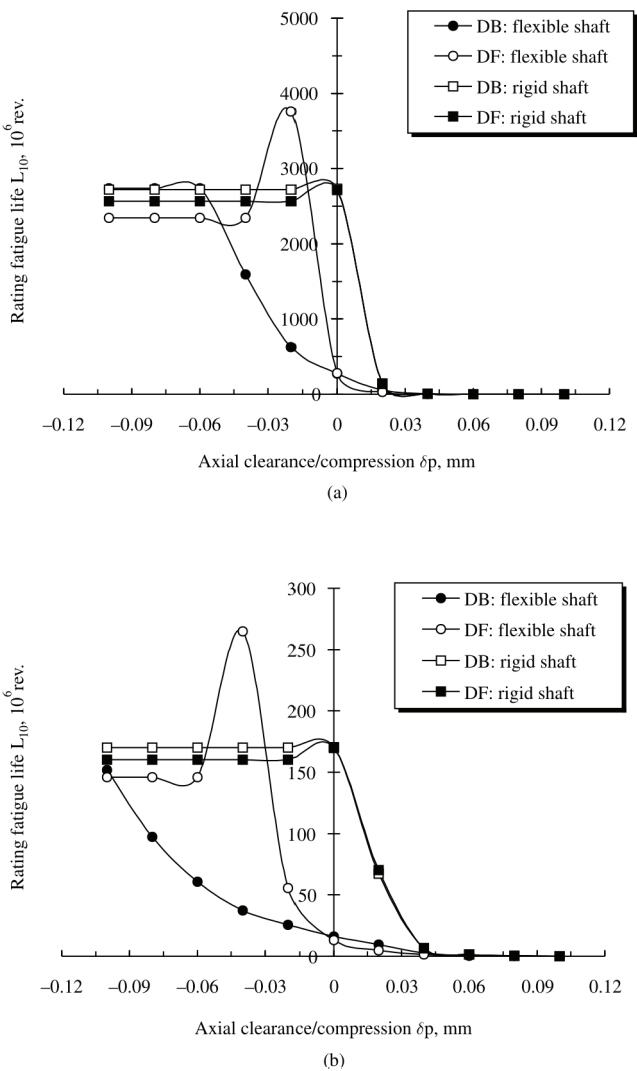


Fig. 9 L_{10} fatigue life versus axial compression: (a) $F_a=3000\text{ N}$; (b) $F_a=F_r=6000\text{ N}$

pression must be judiciously chosen by a compromise between the L_{10} life and the stiffness of the bearing assembly (allowing shaft bending). Also, it can be noted that there is a significant difference between optimum

Table 4 Comparison between bearing loads calculated for flexible and rigid shafts in cantilever design ($\delta_p=0.02\text{ mm}$, $F_a=3000\text{ N}$, $l=354\text{ mm}$ and $c=80\text{ mm}$)

F_r (N)	DTRTB (DB)*				DTRTB (DF)*				DTRTB (DB)†				DTRTB (DF)†			
	Bearing A		Bearing B		Bearing A		Bearing B		Bearing A		Bearing B		Bearing A		Bearing B	
	F_{xA} (N)	F_{yA} (N)	F_{xB} (N)	F_{yB} (N)	F_{xA} (N)	F_{yA} (N)	F_{xB} (N)	F_{yB} (N)	F_{xA} (N)	F_{yA} (N)	F_{xB} (N)	F_{yB} (N)	F_{xA} (N)	F_{yA} (N)	F_{xB} (N)	F_{yB} (N)
3000	1521	3143	1479	−143	1089	3648	1911	−647	1498	3660	1502	−660	1499	3677	1501	−677
6000	1927	6318	1073	−317	−29.6	7354	3030	−1354	1493	7321	1507	−1321	1497	7354	1503	−1354
9000	2776	9534	223	−534	−851	11020	3851	−2019	1484	10980	1516	−1983	1493	11030	1507	−2031

*Considering that the shaft bends.
†Considering that the shaft is infinitely rigid.

preloads (or initial axial clearance/compression) found in the case of flexible or rigid shafts.

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APPENDIX

According to the product law of probability, the survival probability of a system generally containing

two support bearings is equal to the product of the survival probability of each bearing row. Thus, in our example, the following expression can be obtained:

$$\left(\frac{P}{C}\right)^{\omega} = \left(\frac{P_{A1}}{C_{A1}}\right)^{\omega} + \left(\frac{P_{A2}}{C_{A2}}\right)^{\omega} + \left(\frac{P_{B1}}{C_{B1}}\right)^{\omega} + \left(\frac{P_{B2}}{C_{B2}}\right)^{\omega} \quad (11)$$

where C is the basic load rating of the bearing, P is the equivalent applied load and the exponent ω is 3.75 for roller bearings and 3.33 for ball bearings. The subscripts A and B refer to support bearings A and B respectively, and the subscripts 1 and 2 refer to the bearing row.

According to ANSI–AFBMA Standards [9, 10], the equation for predicting L_{10} life for a single-row rolling bearing is

$$L_{10} = \left(\frac{C}{P}\right)^p \quad (12)$$

where p is the load-life exponent ($p = 3$ for ball bearings and $p = 10/3$ for roller bearings).

Rearrangement of equation (11) using equation (12) gives

$$\left(\frac{1}{L_{10}}\right)^e = \left(\frac{1}{L_{10A1}}\right)^e + \left(\frac{1}{L_{10A2}}\right)^e + \left(\frac{1}{L_{10B1}}\right)^e + \left(\frac{1}{L_{10B2}}\right)^e \quad (13)$$

where the exponent e is 10/9 for a point contact and 9/8 for a line contact.

For the inner and outer contact of each bearing row, the L_{10} life, in terms of millions of revolutions, was calculated in accordance with the Lundberg–Palmgren theory [11] including life adjustment factors:

$$L_{10} = a_1 a_2 a_3 \left(\frac{Q_c}{Q_e}\right)^p \quad (14)$$

where Q_c is the basic dynamic capacity, Q_e is the dynamic equivalent load and a_1 , a_2 and a_3 are the appropriate life adjustment factors. The product of the life adjustment factors, $a_1 a_2 a_3$, was assumed to be equal to 1.0 for each of the computer analyses performed.